Why impact also needs energy

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Learning target: Explain why momentum does not tell the whole impact story.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 3, Day 12.*
Both become 3 times as large; p=mv and average force magnitude=change in momentum/time.

02. Worked example: explain the model
Equal momentum arrows; second energy bar twice the first.

03. Read and interpret the model
Second momentum doubles; kinetic energy quadruples.

04. Apply the physics
4 J and 16 J.

05. Explain your reasoning
Thermal energy, deformation, sound, motion of fragments; total energy conserved.

06. Review: connect an earlier idea *Review: Unit 3, Day 3.*
J=72 N·s right; v_f=18 m/s right.

07. Exit check
Factor 9.

08. STRETCH: test the model
Momentum doubles; kinetic energy quadruples. A valid numerical pair supports both.

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01. Warm-up: bring back an earlier idea *Review: Unit 3, Day 12.*
A has 1/3 of B's speed and kinetic energy: $v=p/m$, $K=p^2/(2\ m)$.

02. Worked example: explain the model
Equal momentum arrows; second energy bar twice the first.

03. Read and interpret the model
Second momentum doubles; kinetic energy quadruples.

04. Apply the physics
Speeds 6 and 2 m/s; energies 36 and 12 J.

05. Explain your reasoning
Thermal energy, deformation, sound, motion of fragments; total energy conserved.

06. Review: connect an earlier idea *Review: Unit 3, Day 3.*
$J=12$ N·s; $\Delta v=15$ m/s in the force direction.

07. Exit check
One third.

08. STRETCH: test the model
From $v=p/m$, $K=p^2/(2\ m)$. Fourfold mass gives one quarter the energy at fixed momentum.